AegisAI

Technical Documentation

Models and Algorithms Reference

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1. Introduction

AegisAI is a multi-layered security system designed to detect various cyber threats, including phishing, spam, malicious URLs, and prompt injection attacks. It utilizes a hybrid AI architecture that combines traditional machine learning with state-of-the-art transformer-based deep learning models to achieve high accuracy while maintaining computational efficiency.

2. Core Detection Models

2.1. Multinomial Naive Bayes (MNB)

- Purpose: Content-based classification for spam and phishing emails.
- How it works: A probabilistic classifier based on Bayes' theorem. It calculates the probability of a document belonging to a specific class based on the frequency of terms.
- Why used: Exceptionally fast for both training and inference. Robust baseline for keyword-heavy detection tasks.

2.2. Support Vector Machines (SVM)

- Purpose: Malicious URL and domain classification.
- How it works: SVM finds the optimal hyperplane that maximizes the margin between classes in a high-dimensional feature space.
- Why used: Highly effective in high-dimensional spaces and robust against overfitting.

2.3. DeBERTa-v3 (Transformers)

- Purpose: Deep semantic analysis for phishing and prompt injection.
- How it works: A transformer-based model that uses a disentangled attention mechanism. It understands the context and nuances of text far better than traditional models.
- Specific Usage: Small-scale DeBERTa for phishing; MNLI fine-tuned for prompt injection detection (detecting semantic contradictions).

2.4. DistilBERT

- Purpose: Sentiment and Urgency Analysis.
- Why used: Detects "vulnerability markers" such as extreme urgency or high-pressure language common in phishing attacks.

3. Key Algorithms

3.1. TF-IDF

Application: Feature extraction for MNB and SVM. Weights terms by frequency relative to scarcity across the dataset.

3.2. Sentence-Transformers (MiniLM-L6-v2)

Application: Semantic Similarity. Maps sentences to a dense 384-dimensional vector space to detect brand impersonation (e.g., Homoglyph attacks).

3.3. Cosine Similarity

Logic: Measures the angle between two vectors. High similarity indicates related domains or spoofed brands.

4. Ensemble Architecture

AegisAI uses a Weighted Ensemble Voting system (Synthesizer Agent):
- Phishing Score: 40%
- URL Risk Score: 30%
- Prompt Injection: 30%
- Spam Probability: 15%
- Sentiment Analysis: 10%

The system triggers a high-priority alert if the total weighted risk exceeds 0.65.

5. Summary Table

Model Class	Algorithm	Primary Strength
Traditional ML	Multinomial NB	Ultra-fast baseline detection
Statistical	TF-IDF	Feature relevance ranking
Deep Learning	DeBERTa-v3	Semantic context awareness
Metric Learning	Cosine Similarity	Impersonation detection
Ensemble	Weighted Sum	Consensus-based accuracy